

A two-sided inner intertwining sequence for two states

`MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence`

def

An inner sequence records both forward and actual inverse convergence data before any limiting automorphism is constructed.

Statement

Let A be the completed CAR algebra, the norm completion of the matrix stages $A_n = M_{2^n}(\mathbb{C})$ under $a \mapsto a \otimes I_2$, with the fixed coordinate reindexing. For a unitary $u \in A$, write $\text{Ad}(u)(a) = uau^*$. Given continuous complex-linear functionals $\varphi, \psi : A \rightarrow \mathbb{C}$, an inner intertwining sequence is a sequence (α_n) of complex star automorphisms with the following properties. For every $a \in A$, both $(\alpha_n(a))$ and $(\alpha_n^{-1}(a))$ are Cauchy in norm. Each $\alpha_n = \text{Ad}(u_n)$ for some unitary u_n , and $\varphi(\alpha_n(a)) \rightarrow \psi(a)$ for every $a \in A$.

Definition

Choose the automorphisms first and require the four conditions for those choices. Innerness is expressed as existence of an implementing unitary for each index. Norm completeness supplies pointwise limits from the two Cauchy conditions, while the exact inverse identities allow the two limits to be proved inverse to each other.

Assumptions

Only continuity and complex linearity of the two functionals are inputs to this definition. Positivity, normalization, and purity are not required here. The inverse in the second Cauchy condition is the inverse of that same α_n .

Conclusion

The datum is a sequence with four requirements: forward Cauchy behavior, inverse Cauchy behavior, innerness of each term, and pointwise convergence of the transported functional. A limiting automorphism is a later construction, not an additional field of the datum.

Two unrelated Cauchy sequences do not meet the inverse requirement. No convergence of (u_n) itself is asserted; the relevant convergence is of their conjugation actions on individual algebra elements.

Main citations

- Definition and its exact construction — `MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence`
- Two-sided pointwise limit and its inverse — `MathlibAnnex.CStarAlgebra.twoSidedPointwiseLimit`
- Identification of the inverse limit — `MathlibAnnex.CStarAlgebra.twoSidedPointwiseLimit_symm_apply`

Lean source signature (exact)

```
def HasInnerIntertwiningSequence (phi psi : Limit →L[ $\mathbb{C}$ ]  $\mathbb{C}$ ) : Prop :=
  ∃ f :  $\mathbb{N} \rightarrow \text{StarAlgEquiv } \mathbb{C} \text{ Limit Limit},$ 
    (∀ a, CauchySeq (fun n => f n a)) ∧
    (∀ a, CauchySeq (fun n => (f n).symm a)) ∧
    (∀ n, ∃ u : unitary Limit,
      f n = Unitary.conjStarAlgAut  $\mathbb{C}$  Limit u) ∧
    ∀ a, Tendsto (fun n => phi (f n a)) atTop (nhds (psi a))
```

In the source	Mathematical meaning
<code>phi psi : Limit →L[$\mathbb{C}$] $\mathbb{C}$</code>	The two input continuous complex-linear functionals $\varphi, \psi : A \rightarrow \mathbb{C}$, not necessarily positive, normalized or pure.
<code>∃ f : $\mathbb{N} \rightarrow \text{StarAlgEquiv } \mathbb{C} \text{ Limit Limit}$</code>	There is one sequence $f(n) = \alpha_n$ of complex star automorphisms of the CAR algebra A with all four following conditions.
<code>∀ a, CauchySeq (fun n => f n a)</code>	For each fixed $a \in A$, the sequence $\alpha_n(a)$ is Cauchy in the C^* -norm of A .
<code>∀ a, CauchySeq (fun n => (f n).symm a)</code>	For each fixed $a \in A$, the actual inverse images $\alpha_n^{-1}(a)$ of that same sequence are norm-Cauchy. <code>.symm</code> is not a separately chosen sequence.
<code>∀ n, ∃ u : unitary Limit, f n = Unitary.conjStarAlgAut $\mathbb{C}$ Limit u</code>	For each n , some unitary $u_n \in A$ implements the whole automorphism: $\alpha_n(a) = u_n a u_n^*$ for every $a \in A$.
<code>∀ a, Tendsto (fun n => phi (f n a)) atTop (nhds (psi a))</code>	For each fixed $a \in A$, the complex values $\varphi(\alpha_n(a))$ converge to $\psi(a)$ as $n \rightarrow \infty$. All four conditions concern the one sequence α_n .

Further source notes: The declaration has no pure-state hypotheses.

Earlier published Card and PDF

▼ Exact content identity

Declaration: `MathlibAnnex.CStarAlgebra.CAR.HasInnerIntertwiningSequence`

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Source commit: `437e6e46228bbb8e91211ebded349d7a30020e73`